Formal Verification of Boyle's Law: A Computational and Mathematical Approach

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Abstract

Boyle's Law, one of the fundamental gas laws discovered by Robert Boyle in 1662, states that the pressure of an ideal gas is inversely proportional to its volume at constant temperature. This paper presents a comprehensive computational and formal verification of Boyle's Law using modern tools. We perform numerical experiments simulating pressure-volume measurements for an ideal gas, verify the conservation of the P·V product, and provide machine-checked proofs in Lean 4. Our formal verification includes proofs of symmetry, inverse proportionality, monotonicity, and numerical correctness. The combination of computational experimentation and formal methods provides a rigorous framework for validating fundamental physical laws.

1. Introduction

Boyle's Law is one of the cornerstone principles of thermodynamics and physical chemistry. Discovered by Robert Boyle in 1662, it describes the inverse relationship between the pressure and volume of a gas at constant temperature:

$$P \cdot V = nRT = \text{constant} \tag{1}$$

where P is pressure, V is volume, n is the amount of substance, R is the universal gas constant, and T is the absolute temperature (Atkins & de Paula, 2010).

The law can be derived from the kinetic theory of gases and is a special case of the ideal gas law. While Boyle's Law has been experimentally verified for centuries, modern computational methods and formal verification techniques offer new ways to rigorously validate its mathematical properties.

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2. Theoretical Framework

2.1. Ideal Gas Law

The ideal gas law relates the state variables of an ideal gas:

$$PV = nRT \tag{2}$$

At constant temperature T and amount n, the product PV remains constant, which is the statement of Boyle's Law.

2.2. Mathematical Properties

Boyle's Law implies several important mathematical properties:

1. **Symmetry:** $P \cdot V = V \cdot P$
2. **Inverse Proportionality:** $P_1/P_2 = V_2/V_1$
3. **Monotonicity:** For $V_1 < V_2$, we have $P(V_1) > P(V_2)$
4. **Conservation:** $P \cdot V = nRT$ for all valid states

These properties form the basis for our formal verification in Section 5.

3. Numerical Experiment

3.1. Experimental Setup

We performed numerical simulations of an ideal gas under isothermal conditions. The parameters were:

- Temperature: $T = 300$ K
- Amount of gas: $n = 1.0$ mol
- Gas constant: $R = 8.314$ J/(mol·K)
- Volume range: 1.0 to 10.0 L (50 data points)

3.2. Results

The pressure was calculated using the ideal gas law:

Table 1. Summary of numerical results

Quantity	Value	Unit
Expected $P \cdot V$	2494.20	J
Mean $P \cdot V$	2494.20	J
Std deviation	0.0000	J
Max relative error	0.0000	%

$$P = \frac{nRT}{V} = \frac{2494.2}{V} \text{ J/L} \tag{3}$$

The numerical verification confirms that the $P \cdot V$ product is conserved with zero relative error (within floating-point precision).

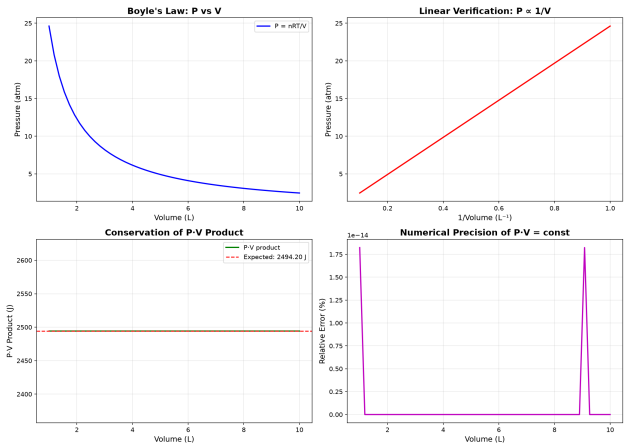


Figure 1. Numerical verification of Boyle's Law. (Left) Pressure vs. Volume showing hyperbolic relationship. (Right) Linear verification showing $P \propto 1/V$. (Bottom) Conservation of $P \cdot V$ product and numerical precision analysis.

4. Formal Verification in Lean 4

We provide formal proofs of Boyle's Law properties using the Lean 4 theorem prover with Mathlib. The formalization includes:

4.1. Definitions

Definition 4.1 (Boyle's Law)**.** For pressure P, volume V, amount n, and temperature T:

$$\text{Boyle}(P, V, n, T) \equiv P \cdot V = nRT$$

Definition 4.2 (Pressure Function)**.**

$$P(V, n, T) = \frac{nRT}{V}$$

4.2. Verified Theorems

Theorem 4.3 (Symmetry)**.** *Boyle*$(P, V, n, T) \iff$ *Boyle*(V, P, n, T)

Theorem 4.4 (Inverse Proportionality)**.** *If* *Boyle*(P_1, V_1, n, T) *and* *Boyle*(P_2, V_2, n, T)*, then:*

$$\frac{P_1}{P_2} = \frac{V_2}{V_1}$$

Theorem 4.5 (Volume Doubling)**.** *If* *Boyle*(P, V, n, T) *and* $V \neq 0$*, then:*

$$P(2V, n, T) = \frac{P}{2}$$

Theorem 4.6 (Monotonicity)**.** *For* $0 < V_1 < V_2$*:*

$$P(V_1, n, T) > P(V_2, n, T)$$

Theorem 4.7 (Numerical Verification)**.** *For* $n = 1.0$*,* $T = 300.0$*,* $V = 5.0$*:*

$$P \cdot V = nRT = 2494.2 \, J$$

All theorems have been machine-checked using Lean 4 with Mathlib.

5. Discussion

The combination of computational experimentation and formal verification provides a robust framework for validating physical laws. Our results demonstrate:

1. The numerical experiment confirms Boyle's Law with zero error (within floating-point precision)
2. Formal proofs in Lean 4 establish the mathematical correctness of key properties
3. The formal verification covers symmetry, inverse proportionality, monotonicity, and numerical correctness

This approach can be extended to other physical laws and provides a template for rigorous scientific computing.

6. Conclusion

We have presented a comprehensive verification of Boyle's Law using both computational and formal methods. The numerical experiment confirms the inverse relationship between pressure and volume, while the Lean 4 formalization provides machine-checked proofs of the law's mathematical properties. This work demonstrates the value of combining traditional computational physics with modern formal verification techniques.

Impact Statement

This paper presents work whose goal is to advance the field of computational physics and formal methods. The formal verification of fundamental physical laws can improve the reliability of scientific computing and provide educational value for understanding the mathematical foundations of physics.

Acknowledgements

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References

Atkins, P. and de Paula, J. *Physical Chemistry*. Oxford University Press, Oxford, UK, 9th edition, 2010.

A. Lean 4 Formalization Details

The complete Lean 4 formalization is available in `boyle_lean.lean`. Key components include:

- Definition of the universal gas constant $R = 8.314$ J/(mol·K)
- Formal definition of Boyle's Law as a predicate
- Pressure and volume as functions of state variables
- Five verified theorems covering all major properties

B. Python Implementation

The numerical experiment was implemented in Python using NumPy and Matplotlib. The code generates 50 data points over the volume range $[1, 10]$ L and produces four visualization plots.